

**Unit ID: 6798d446aaf593b2823f1981**

## **Unit 5 - Knowledge Check**

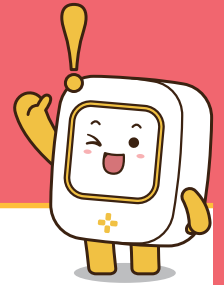
### **PDF Documents**

**PDF Document 1: 1738069060942-Chapter 4 - Unit 5 - Knowledge Check.pdf**

This PDF document is included in the unit materials.

**Update #1 - 2025-05-16 15:19:45**

# KNOWLEDGE CHECK



**1** What happens when you combine red and blue light?

- ☐ A The light disappears.
- ☐ B It creates green light.
- ☐ C It creates magenta light.
- ☐ D It creates yellow light.

CORRECT ANSWER

☒ C

**2** Which primary color of light has the longest wavelength?

- ☐ A Blue
- ☐ B Green
- ☐ C Red
- ☐ D Yellow

CORRECT ANSWER

☒ C

**3** What color is produced when green and blue light are mixed?

- ☐ A Red
- ☐ B Cyan
- ☐ C Magenta
- ☐ D Yellow

CORRECT ANSWER

☒ B

**4** If all three primary colors (RGB) are mixed in equal measure, what color is produced?

- ☐ A Black
- ☐ B White
- ☐ C Purple
- ☐ D Orange

CORRECT ANSWER

☒ B

**5** What is the purpose of using variables in programming the RGB LED module?

- ☐ A To store color values that can change during execution.
- ☐ B To increase the brightness of the LED.
- ☐ C To connect the LED to the internet.
- ☐ D To create a delay between color changes.

CORRECT ANSWER

☒ A

**6** Which of the following is a true statement about 'nested if' statements in programming?

- ☐ A They can only be used for error checking.
- ☐ B They allow for multiple conditions to be checked sequentially within another condition.
- ☐ C They reduce the efficiency of the program and should be avoided.
- ☐ D They are used to repeat a block of code multiple times.

CORRECT ANSWER

☒ B

**7** What should you do if an RGB value variable accidentally goes negative due to a programming error?

- ☐ A Reset the variable to zero.
- ☐ B Increase the value by 100.
- ☐ C Ignore the error.
- ☐ D Turn off the LED.

CORRECT ANSWER

☒ A

## Critical Thinking Challenge

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- 1** Consider how the lighting in a room affects your mood or productivity. Write a brief paragraph explaining how you might use RGB LEDs to enhance the lighting of your study area to better suit your learning environment.

- 2** Think about the implications of using RGB lighting in public spaces. Write a brief paragraph discussing how changing the color of lights in places like parks, theaters, or public transport can affect the mood and behavior of people in those spaces.